Food Recognition with Transfer Learning (Prodigy InfoTech – Task 05)

Machine Learning internship at **Prodigy InfoTech**, Task 05.

> **Task:** Develop a model that can accurately recognize food items from images and estimate their calorie content, enabling users to track their dietary intake and make informed food choices.

Dataset

Food-101 (ETH Zurich) has **101,000 images** across **101 food categories**, with 1,000 images per class. The notebook downloads and extracts it automatically.

Split	Images
Training (80%)	80,800
Validation (20%)	20,200

Approach

1. **Data pipeline.** `ImageDataGenerator` with rescaling and 80/20 train/validation split.
2. **Data augmentation**
   - Rotation (±20°).
   - Width and height shift (0.2).
   - Shear and zoom (0.2).
   - Horizontal flip.
3. **Transfer learning**
   - **InceptionV3** pre-trained on ImageNet, used as a frozen feature extractor (input 299 × 299 × 3).
   - Custom head: `GlobalAveragePooling2D → Dense(1024, ReLU) → Dense(101, softmax)`.
   - 24.0 M parameters in total, of which **2.2 M are trainable**.
4. **Training.** Adam optimizer with categorical cross-entropy. A faster experiment uses 150 × 150 images and a limited number of batches.

Tech stack

Python · TensorFlow / Keras · InceptionV3 · NumPy · Matplotlib · Google Colab (GPU)

How to run

1. Open `Prodigy_ML_05.ipynb` in Google Colab with a GPU runtime.
2. Run the first cell to download and extract Food-101 (about 5 GB).
3. Run the remaining cells to build and train the model.

Next steps

- Train for more epochs at full resolution, then fine-tune the top Inception blocks.
- Add a calorie lookup table that maps each of the 101 classes to typical calories per serving, so a recognized dish returns a calorie estimate.

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